OOPJ DAILY ASSIGNMENT – 9 (12/01/26)

Name – Nishant Nahar

Roll No – 241551078

1. Simple Java Package

```
package mypack;
public class Simple {
    public static void main(String args[]) {
        System.out.println("Welcome to package");
    }}
```

OUTPUT

```
B:\java_theory\assignment_7  javac -d . Simple.java
B:\java_theory\assignment_7  java mypack.Simple
Welcome to package
```

2. Using packagename.*
   a. A Pack

```
package pack;
public class A {
    public void msg() {
        System.out.println("Hello");
    }}
```

   b. B Pack

```
package mypack_1;
import pack.*;
class B {
    public static void main(String args[]) {
        A obj = new A();
        obj.msg();
    }
}
```

OUTPUT

```
B:\java_theory\assignment_7  javac -d . A.java
B:\java_theory\assignment_7  javac -d . B.java
B:\java_theory\assignment_7  java mypack_1.B
Hello
```

3. Using packagename.classname
   a. A Pack

```
package pack_1;
public class A_1 {
    public void msg() {
        System.out.println("Hello");
    }
```

```
}
```

b. B Pack

```
package mypack_2;
import pack_1.A_1;
class B_1 {
    public static void main(String args[]) {
        A_1 obj = new A_1();
        obj.msg();
    }
}
```

OUTPUT

```
B:\java_theory\assignment_7 > javac -d . A_1.java
B:\java_theory\assignment_7 > javac -d . B_1.java
B:\java_theory\assignment_7 > java mypack_2.B_1
Hello
```

4. Using fully qualified name
    a. A Pack

```
package pack_2;
public class A_2 {
    public void msg() {
        System.out.println("Hello");
    }
}
```

b. B Pack

```
package mypack_3;
class B_2 {
    public static void main(String args[]) {
        pack_2.A_2 obj = new pack_2.A_2();// using fully qualified name
        obj.msg();
    }
}
```

OUTPUT

```
B:\java_theory\assignment_7 > javac -d . A_2.java
B:\java_theory\assignment_7 > javac -d . B_2.java
B:\java_theory\assignment_7 > java mypack_3.B_2
Hello
```

5. Private Access Modifier

```
class pvt {
    private int data = 40;
    private void msg() {
        System.out.println("Hello java");
    }
}
public class pvt_1 {
    public static void main(String args[]) {
        pvt obj = new pvt();
```

```
        System.out.println(obj.data);// Compile Time Error
        obj.msg();// Compile Time Error
    }
}
```

OUTPUT

```
B:\java_theory\assignment_7  java pvt_1.java
pvt_1.java:14: error: data has private access in pvt
        System.out.println(obj.data);// Compile Time Error
                           ^
pvt_1.java:15: error: msg() has private access in pvt
        obj.msg();// Compile Time Error
           ^
2 errors
error: compilation failed
```

6. Private Modifier Role

```
class A_4 {
    private A_4() {
    }// private constructor
    void msg() {
        System.out.println("Hello java");
    }
}
public class pvt_role {
    public static void main(String args[]) {
        A_4 obj = new A_4();// Compile Time Error
    }
}
```

OUTPUT

```
B:\java_theory\assignment_7  java .\pvt_role.java
.\pvt_role.java:16: error: A_4() has private access in A_4
        A_4 obj = new A_4();// Compile Time Error
                  ^
1 error
error: compilation failed
```

7. Default Access Modifier
    a. A Pack

```
package pack_3;
class def_A {
    void msg() {
        System.out.println("Hello");
    }
}
```

    b. B Pack

```
package mypack_4;
import pack_3.*;
class def_B {
    public static void main(String args[]) {
        def_A obj = new def_A();// Compile Time Error
        obj.msg();// Compile Time Error
```

```
    }
}
```

OUTPUT

```
B:\java_theory\assignment_7  javac -d . def_A.java
B:\java_theory\assignment_7  javac -d . def_B.java
def_B.java:7: error: cannot find symbol
        def_A obj = new def_A();// Compile Time Error
        ^
  symbol:   class def_A
  location: class def_B
def_B.java:7: error: cannot find symbol
        def_A obj = new def_A();// Compile Time Error
                        ^
  symbol:   class def_A
  location: class def_B
2 errors
```

8. Protected Access Monitor
    a. A Pack

```
package pack_4;
public class pro_A {
    protected void msg() {
        System.out.println("Hello");
    }}
```

    b. B Pack (with extends)

```
package mypack_4;
import pack_4.*;
class pro_B extends pro_A {
    public static void main(String args[]) {
        pro_B obj = new pro_B();
        obj.msg();
    }}
```

OUTPUT

```
B:\java_theory\assignment_7  javac -d . pro_A.java
B:\java_theory\assignment_7  javac -d . pro_B.java
B:\java_theory\assignment_7  java mypack_4.pro_B
Hello
```

    c. B Pack (Without Extends)

```
package mypack_4;
import pack_4.*;
class pro_B{
    public static void main(String args[]) {
        pro_B obj = new pro_B();
        obj.msg();
    }}
```

OUTPUT

```
B:\java_theory\assignment_7 javac -d . pro_A.java
B:\java_theory\assignment_7 javac -d . pro_B.java
pro_B.java:8: error: cannot find symbol
        obj.msg();
          ^
  symbol:   method msg()
  location: variable obj of type pro_B
1 error
```

9. Public Access Modifier
    a. A Pack

```java
package pack_5;
public class pub_A {
    public void msg() {
        System.out.println("Hello");
    }}
```

    b. B Pack

```java
package mypack_5;
import pack_5.*;
class pub_B {
    public static void main(String args[]) {
        pub_A obj = new pub_A();
        obj.msg();
    }}
```

OUTPUT

```
B:\java_theory\assignment_7 javac -d . pub_A.java
B:\java_theory\assignment_7 javac -d . pub_B.java
B:\java_theory\assignment_7 java mypack_5.pub_B
Hello
```

10. Method Overridding

```java
class A_3 {
    protected void msg() {
        System.out.println("Hello java");
    }}
public class meth_over extends A_3 {
    void msg() {
        System.out.println("Hello java");
    }   // C.T.Error
    public static void main(String args[]) {
        meth_over obj = new meth_over();
        obj.msg();
    }}
```

OUTPUT

```
B:\java_theory\assignment_7 java .\meth_over.java
.\meth_over.java:8: error: msg() in meth_over cannot override msg() in A_3
    void msg() {
        ^
  attempting to assign weaker access privileges; was protected
1 error
error: compilation failed
```